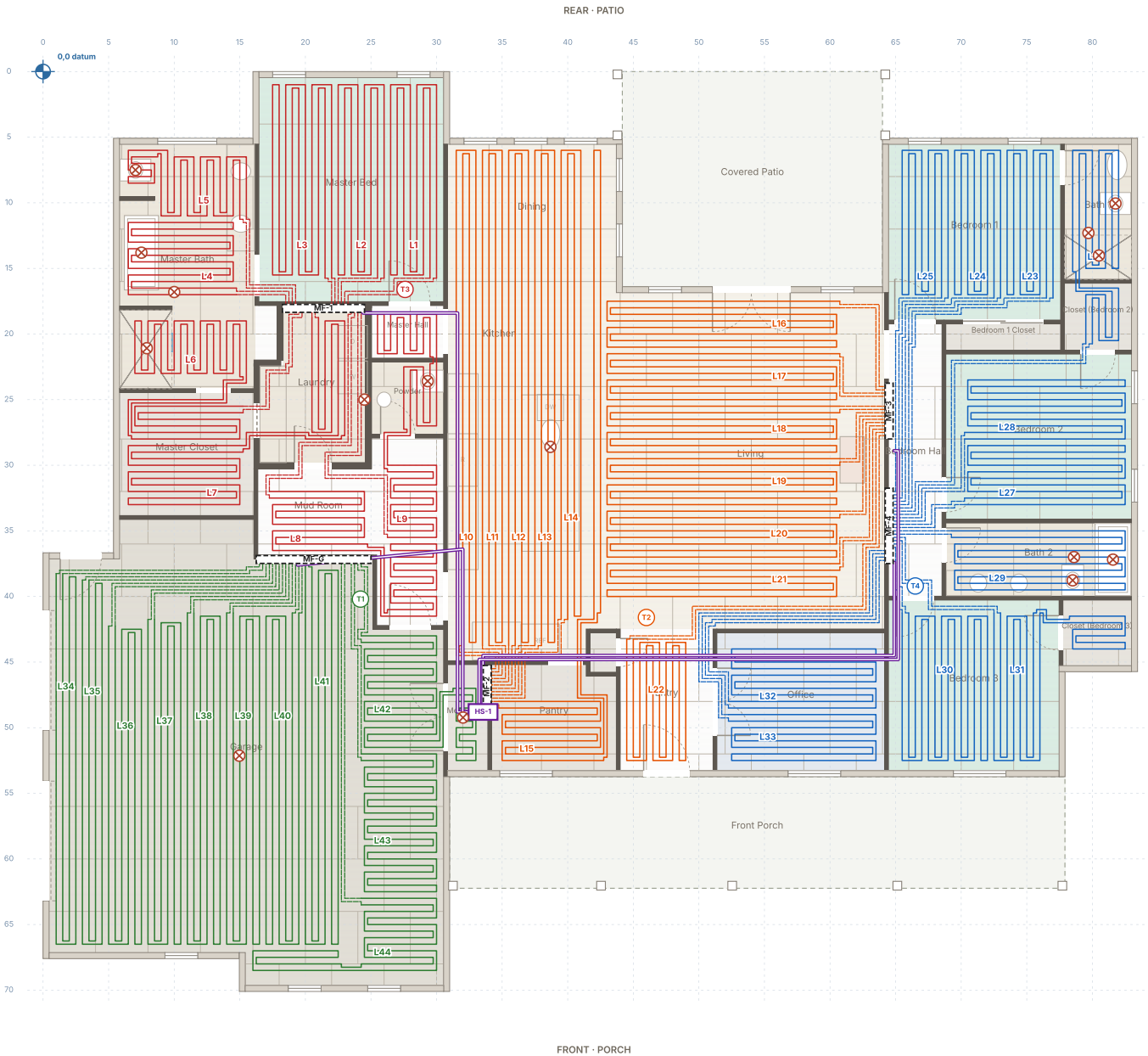


Zone 1, Garage (held at 50°F) (T1)

Zone 2, Great room (T2)

Zone 3, Master suite and utility (T3)

Zone 4, Bedroom wing and office (T4)

----- leaders, supply and return side by side 3" apart

1" feed pair from the closet circulators, under the boards

manifold cabinet over its row of 3" ports

4' x 4' nub boards in a running bond from the datum; dots are the 3" puck grid

44 loops, 7,200' of 1/2" PEX, 5 manifolds, no loop over 200'. · Green zone 1 (garage), orange zone 2, red zone 3, blue zone 4. · 54 kBtu/h at design, 110°F supply. · EPS25 Super 4 nub boards on the 3" grid from the datum; nothing crosses; tube under every fixture.

draft

R-1 Radiant · 64 Stone Sheep Circle

PEX loops on 3" nub boards under the slab: 200' loops, nothing crossing, five manifolds on four zones with a circulator each, flows at Powell, WY design (-11°F). App version v28. From the owner's plans (Larson House, 6/11/26).

Notes

1. The layout is drawn on the boards' own grid. EPS25 Super 4 nub boards (4' x 4', 5" thick: 4" of EPS, R-18, under a 1" grid of pucks on 3" centres) go down in a running bond with their puck rows on the 3" lines from the datum at 0,0 (the garage-door wall's outside face and the master bed's rear wall line, as on every sheet), so every tube on this sheet sits in a channel between pucks. Lay the first board's corner at the datum and the rest follow; a board cut at a wall keeps its grid.
2. Tube: 44 loops of 1/2" O2-barrier PEX, none over 200' with its leaders and the 2' rises at the cabinet (L6 is the longest at 192', L5 the shortest at 116'). Fields at 6" o.c. (every other channel), supply and return side by side so each loop's two ends leave its field together; leaders run as 3" pairs through doorways and along walls to their manifold. Nothing crosses anything: the whole layout was checked tube by tube on the grid. One continuous coil per loop, no fittings in the slab; tag both ends with the loop number at the cabinet. The loops on one manifold are planned to like lengths so they balance on their flow meters (MF-1 116'-192', MF-2 167'-189', MF-3 143'-173', MF-4 136'-181', MF-G 147'-187'); MF-1's spread is the master bath, whose three small fields are boxed in by the toilet and shower walls and can neither join each other nor share rows with the bedroom.
3. Under a wall only where drawn, straight across and only for the length drawn: under the master bed wall into MF-1, under the kitchen wall into the pantry and under the pantry wall into MF-2, under the living room wall into MF-3 and MF-4, under the master hall wall into the powder room, and under the bath 1 wall from the bedroom 2 closet. Mark every wall crossing on the boards before the pour so the plate anchors and shots stay 12" off it; everywhere else the leaders go through the doorway.
4. The tube runs under everything that sits on the floor: the kitchen base cabinets, island, fridge and range, the washer and dryer, the tubs, showers, vanities and toilets, the hearth. It lies about 5" below the top of the slab, so nothing fastened to the floor reaches it; only the walls

Open items

- L31 (Bedroom 3, Closet (Bedroom 3)): the floor carries all but about 345 Btu/h of the room at -11°F with 110°F water. Add a small electric towel warmer, or run the boiler hotter on the coldest days.
- Not tubed: the coat closet (about 6 sq ft), the bedroom 1 bypass closet (about 20 sq ft), the 3' x 10' notch at the garage's north wall beside the master closet (about 31 sq ft, left bare by decision 2026-10-14), the laundry's west floor along the counter (about 15 sq ft: every way of tubing it either takes the laundry loop past 200' or crosses the closet and mud room leaders that use that floor) and the west 18" of the powder room under the lav (about 9 sq ft: the powder is the middle panel of the mud room leg's loop, which can only be 3' or 5' wide there, and 5' would not clear the east wall). The first three are too small for a loop of their own at the lengths on their manifold, and none can be added to a neighbouring loop without crossing another. All five sit inside heated floor and borrow from it.
- Boiler model: HS-1 is a condensing boiler on natural gas or propane (decided 2026-10-13). Pick a combi or a boiler with an indirect tank if it should also make the domestic hot water and replace P-1's 50 gal heater; propane needs the boiler's LP conversion kit.
- Garage: kept at 50°F on its own manifold, pump and thermostat. Decide whether it runs a glycol mix behind a plate heat exchanger, which protects it if the house loses power in a cold snap with the doors open.
- Flooring: the loops assume hard-surface floors. Carpet in a bedroom needs about 10°F warmer water for that loop.
- S-1: MF-3 and MF-4 recess into the hall's 2×4 west wall (3-1/2" deep; fur locally if the chosen manifold needs more). MF-4's recess is 2'-0" longer than S-1 framed it, 32'-0" to 37'-3" along the wall; the MF-K and MF-5 recesses are no longer needed and MF-1's surface cabinet now runs 18'-6" to 24'-3" at 4' up, over the dryer.
- E-1: no actuators, so the receptacles at the manifold cabinets are not needed; HS-1, the four circulators and the zone controller stay on circuit 27, and T1 to T4 wire to the zone controller in the closet.

and their plates, the posts and the manifold port rows keep it out. Where P-1 brings a drain, a tub trap, a closet flange or a vent stack up through the slab (15 places, each marked with a crossed circle) shift the one run it meets over a channel for the length of the fitting, so the tube stays 3" clear of the flange or drain body; the showers' floors are heated like the rest.

5. Manifolds, five stations, each wholly on one zone, each with a 3" port position per tube end in one row along its wall (the schedule gives every loop's two port numbers, counted from the low end of the row): MF-1 (laundry, on the north wall from the linen alcove to the east wall, over the dryer, 9 loops) serves Master Bed, Master Bath, Master Closet, Master Hall, Laundry, Powder, Mud Room; MF-2 (mechanical closet, low on the east wall under HS-1, 6 loops) serves Kitchen, Pantry; MF-3 (bedroom hall, west wall north of the fireplace, 7 loops) serves Living, Entry; MF-4 (bedroom hall, west wall south of the fireplace, 11 loops) serves Office, Bedroom 1, Bath 1, Closet (Bedroom 2), Bedroom 2, Bath 2, Bedroom 3, Closet (Bedroom 3); MF-G (garage, on the mud room wall (two 7-loop manifolds side by side), 11 loops) serves Mechanical, Garage. The four circulators are in the mechanical closet with HS-1: MF-2 is piped to P-R2 there; MF-1, MF-3, MF-4 and MF-G each get a 1" insulated pair under the boards, 162' each way in all.
6. The leaders rise straight up out of the slab at their port positions into the cabinet (sleeve each rise in 12" of pipe insulation where it leaves the concrete) and bundle up to the manifold; they may be gathered sideways in the last 12" below the cabinet. Cabinets are recessed in the hall wall (MF-3, MF-4) and surface-mounted on the laundry's sound wall (MF-1), low on the closet's east wall under the boiler (MF-2) and on the garage face of the mud room wall (MF-G).
7. Zoning is by manifold: four zones (garage; great room; master suite and utility; bedroom wing and office), each a circulator in the mechanical closet started by its thermostat, and every loop on a manifold is on that manifold's zone. Zone 2 has two manifolds, MF-2 and MF-3, on the one circulator. No actuators: balance each loop on its flow meter.
8. Flow: 10.9 gpm with every loop calling, each loop set on its flow meter at a 10°F drop. Design heat 53,534 Btu/h, of which 10,924 is the garage at 50°F; 7% is added for heat lost down through the R-18 boards and out the slab edge. Rooms with no field of their own are heated by the leader bundles running through them: Dining (231' of

- P-1: the closet's east wall is shared, MF-2 low at its north end under HS-1 and the water heater at its south end; the main water manifold and its risers stay west of MF-2's port column (x 33'-6"). The pantry floor carries MF-2's leaders at 3" across its west end and runs warmer than the rooms around it; sleeve the bundle if the pantry should stay cool.
- The EPS25 Super 4 nub boards come 5 to a bundle (80 sq ft), 3 bundles minimum: 55 bundles cover the 4,167 sq ft slab with 5% for cuts. Confirm the lead time; the board layout can start at any corner as long as the grid holds the datum.

leaders for 3,771 Btu/h), Bedroom Hall (166' of leaders for 422 Btu/h). Nothing heats the Coat Closet (30 Btu/h), Bedroom 1 Closet (86 Btu/h); they borrow from the rooms around them.

9. Water: 110°F supply at -11°F outside, reset lower as it warms, so a condensing boiler stays condensing. Floor surfaces stay at or under 100°F. At that supply L31 (Bedroom 3, Closet (Bedroom 3)) falls about 345 Btu/h short at design; see the open items.
10. Pumps: one constant-ΔP ECM circulator per zone, sized on the zone's hardest loop plus about 3 ft for the feed pair and manifold: P-R1 about 4 ft (L36, 2.3 gpm); P-R2 about 5 ft (L15, 3.1 gpm); P-R3 about 5 ft (L6, 2.2 gpm); P-R4 about 5 ft (L26, 3.4 gpm).
11. Under the boards: 10-mil vapor barrier over the compacted base, lapped and taped; R-10 (2") rigid foam at the slab edge. The 4" slab goes over the pucks with the tube tops about 1" below the surface of the boards; no mesh ties are needed, the pucks hold the tube. Pressure test every loop at 100 psi water or 60 psi air for 24 hours before the pour and keep it under pressure through the pour. Photograph every room with a tape in frame before concrete.
12. HS-1, the four circulators and the zone controller are circuit 27 on E-1 (200 A panel in the garage); the manifold cabinets need no power. T1 to T4 wire to the zone controller in the closet.
13. P-1's water lines, vents and drains are all in the base course below the boards and the feed pairs (cold PEX 1'-0" down, hot 1'-1 1/2", drains below both), so the boards go down whole, with no channels cut for plumbing.

Loop schedule

Ports are the supply and return positions counted from the low end of the manifold's row. Leader is each way. Heat includes the share lost downward. Pick a row to find the loop.

Loop	Zone	Rooms	Manifold	Ports	In field	Leader	Total	Btu/h	Flow	Head	Floor
L1	3	Master Bed	MF-1	20/19	121'	11'	144'	959	0.20 gpm	0.3 ft	76°F
L2	3	Master Bed	MF-1	18/17	121'	8'	137'	911	0.20 gpm	0.3 ft	76°F
L3	3	Master Bed	MF-1	16/15	152'	5'	162'	1,083	0.22 gpm	0.4 ft	76°F
L4	3	Master Bath	MF-1	1/2	104'	7'	120'	1,376	0.28 gpm	0.5 ft	81°F
L5	3	Master Bath, Master Bath	MF-1	4/3	90'	13'	116'	1,328	0.27 gpm	0.5 ft	81°F
L6	3	Master Closet, Master Bath	MF-1	5/6	163'	14'	192'	1,870	0.37 gpm	1.4 ft	81°F

Loop	Zone	Rooms	Manifold	Ports	In field	Leader	Total	Btu/h	Flow	Head	Floor
L7	3	Laundry, Master Closet	MF-1	9/10	164'	3'	170'	1,218	0.24 gpm	0.6 ft	78°F
L8	3	Mud Room, Mud Room	MF-1	22/21	140'	23'	186'	1,074	0.21 gpm	0.5 ft	76°F
L9	3	Mud Room, Powder, Master Hall	MF-1	23/24	128'	26'	182'	882	0.20 gpm	0.4 ft	76°F
L10	2	Kitchen	MF-2	1/2	152'	8'	169'	1,207	0.24 gpm	0.6 ft	86°F
L11	2	Kitchen	MF-2	3/4	152'	7'	168'	1,217	0.24 gpm	0.6 ft	86°F
L12	2	Kitchen	MF-2	5/6	152'	7'	167'	1,231	0.25 gpm	0.6 ft	86°F
L13	2	Kitchen	MF-2	7/8	152'	10'	172'	1,265	0.25 gpm	0.6 ft	86°F
L14	2	Kitchen	MF-2	9/10	144'	14'	173'	1,294	0.26 gpm	0.6 ft	86°F
L15	2	Pantry, Kitchen	MF-2	11/12	181'	3'	189'	1,781	0.36 gpm	1.2 ft	86°F
L16	2	Living	MF-3	1/2	145'	12'	171'	995	0.20 gpm	0.4 ft	76°F
L17	2	Living	MF-3	3/4	145'	9'	164'	963	0.20 gpm	0.4 ft	76°F
L18	2	Living	MF-3	7/8	145'	6'	158'	929	0.20 gpm	0.4 ft	76°F
L19	2	Living	MF-3	9/10	145'	10'	165'	973	0.20 gpm	0.4 ft	76°F
L20	2	Living	MF-3	11/12	145'	13'	172'	1,016	0.20 gpm	0.4 ft	76°F
L21	2	Living	MF-3	13/14	108'	17'	143'	839	0.20 gpm	0.3 ft	76°F
L22	2	Entry	MF-3	15/16	96'	38'	173'	1,516	0.30 gpm	0.8 ft	80°F
L23	4	Bedroom 1	MF-4	6/5	93'	30'	154'	1,409	0.28 gpm	0.7 ft	80°F
L24	4	Bedroom 1	MF-4	4/3	93'	25'	145'	1,308	0.26 gpm	0.5 ft	80°F
L25	4	Bedroom 1	MF-4	2/1	93'	21'	136'	1,225	0.25 gpm	0.5 ft	80°F
L26	4	Closet (Bedroom 2), Bath 1	MF-4	7/8	110'	30'	171'	2,423	0.48 gpm	1.9 ft	87°F
L27	4	Bedroom 2	MF-4	12/11	126'	9'	146'	1,091	0.22 gpm	0.4 ft	78°F
L28	4	Bedroom 2	MF-4	10/9	126'	14'	155'	1,155	0.23 gpm	0.5 ft	78°F
L29	4	Bath 2	MF-4	13/14	136'	7'	151'	892	0.20 gpm	0.4 ft	76°F
L30	4	Bedroom 3	MF-4	18/17	164'	8'	181'	2,059	0.41 gpm	1.5 ft	81°F
L31	4	Bedroom 3, Closet (Bedroom 3)	MF-4	16/15	128'	16'	160'	2,365	0.47 gpm	1.7 ft	100°F
L32	4	Office	MF-4	21/22	116'	29'	175'	1,466	0.29 gpm	0.8 ft	79°F
L33	4	Office	MF-4	20/19	93'	31'	157'	1,291	0.26 gpm	0.6 ft	79°F
L34	1	Garage	MF-G	1/2	114'	18'	151'	866	0.20 gpm	0.4 ft	56°F
L35	1	Garage	MF-G	3/4	112'	17'	147'	842	0.20 gpm	0.3 ft	56°F
L36	1	Garage	MF-G	5/6	148'	19'	187'	1,075	0.21 gpm	0.5 ft	56°F
L37	1	Garage	MF-G	7/8	150'	16'	184'	1,057	0.21 gpm	0.5 ft	56°F
L38	1	Garage	MF-G	9/10	154'	13'	181'	1,040	0.21 gpm	0.5 ft	56°F

Loop	Zone	Rooms	Manifold	Ports	In field	Leader	Total	Btu/h	Flow	Head	Floor
L39	1	Garage	MF-G	11/12	154'	11'	176'	1,010	0.20 gpm	0.4 ft	56°F
L40	1	Garage	MF-G	14/13	154'	6'	167'	957	0.20 gpm	0.4 ft	56°F
L41	1	Garage	MF-G	15/16	174'	2'	180'	1,037	0.21 gpm	0.5 ft	56°F
L42	1	Garage, Mechanical	MF-G	32/31	148'	9'	166'	1,102	0.22 gpm	0.5 ft	60°F
L43	1	Garage	MF-G	30/29	136'	17'	172'	990	0.20 gpm	0.4 ft	56°F
L44	1	Garage, Garage	MF-G	27/26	106'	29'	165'	948	0.20 gpm	0.4 ft	56°F
Zone 1	1	Garage (held at 50°F)	MF-G		11 loops		1,876'	10,924	2.3 gpm		
Zone 2	2	Great room	MF-2, MF-3		13 loops		2,184'	15,225	3.1 gpm		
Zone 3	3	Master suite and utility	MF-1		9 loops		1,410'	10,702	2.2 gpm		
Zone 4	4	Bedroom wing and office	MF-4		11 loops		1,730'	16,684	3.4 gpm		

Manifolds

Each station's row of 3" port positions along its wall; a loop takes two next to each other.

Tag	Loops	Ports used	Where	Zones	Feed
MF-1	9 loops: L1, L2, L3, L4, L5, L6, L7, L8, L9	18 of 24	Laundry, on the north wall from the linen alcove to the east wall, over the dryer	3	1" pair F-1, 40' each way, from P-R3
MF-2	6 loops: L10, L11, L12, L13, L14, L15	12 of 12	Mechanical closet, low on the east wall under HS-1	2	piped direct to P-R2 in the closet
MF-3	7 loops: L16, L17, L18, L19, L20, L21, L22	14 of 16	Bedroom hall, west wall north of the fireplace	2	1" pair F-3, 52' each way, from P-R2
MF-4	11 loops: L23, L24, L25, L26, L27, L28, L29, L30, L31, L32, L33	22 of 22	Bedroom hall, west wall south of the fireplace	4	1" pair F-4, 43' each way, from P-R4
MF-G	11 loops: L34, L35, L36, L37, L38, L39, L40, L41, L42, L43, L44	22 of 34	Garage, on the mud room wall (two 7-loop manifolds side by side)	1	1" pair F-G, 28' each way, from P-R1

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The installer takes off the final list.

Item	Qty	Notes
1/2" O2-barrier PEX, 1,000' coils	8 coils	7,200' in 44 uncut lengths of 116' to 192' (the schedule), cut longest first
EPS25 Super 4 nub board, 4' x 4'	274 boards (55 bundles of 5)	4,167 sq ft of slab + 5% for cuts; R-18, 1/2" tube channels on 3" centres
1" PEX supply + return, insulated	341 lf	HS-1 to the stations, in the base course under the boards

Item	Qty	Notes
Stainless manifolds with flow meters	5 stations	MF-1 9-loop, MF-2 6-loop, MF-3 7-loop, MF-4 11-loop, MF-G 11-loop (MF-G as two manifolds side by side)
Zone controller + thermostats	1 + 4	T1 to T4, one per zone, each starting its circulator; outdoor reset sensor
Circulators	4	P-R1 to P-R4, ECM on constant ΔP, one per zone, all in the mechanical closet; P-R1 with the garage mixing valve
Manifold cabinets	5	recessed at MF-3 and MF-4; surface at MF-1, MF-2 and MF-G
R-10 rigid foam, 2", slab edge	346 lf	the perimeter of the slab, full depth
10-mil vapor barrier	4,584 sf	under the boards, lapped and taped
Pipe insulation, 1/2" x 12"	88 pcs	one sleeve on every tube end where it rises out of the concrete